

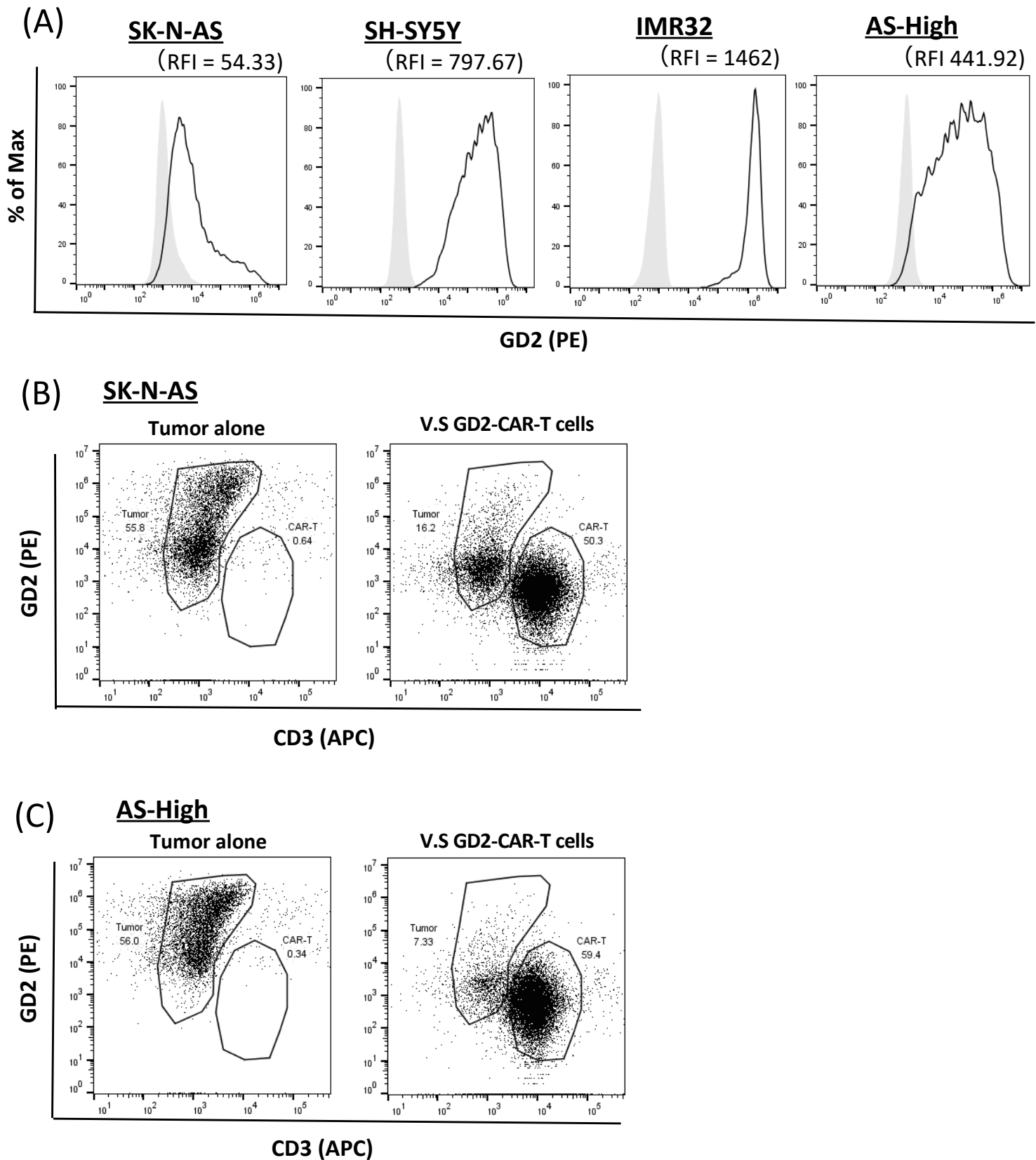


Figure S1 GD2 expression on neuroblastoma (NB) cell lines and coculture assay.

(A) GD2 expression on NB cell lines determined by flow cytometry. Relative fluorescence intensity (RFI) data are shown for each cell line. (B) (C) Representative flow cytometry dot plots of 4-day coculture assay using PB-GD2-CAR-T cells and tumors, SK-N-AS (B) or AS-High (C).

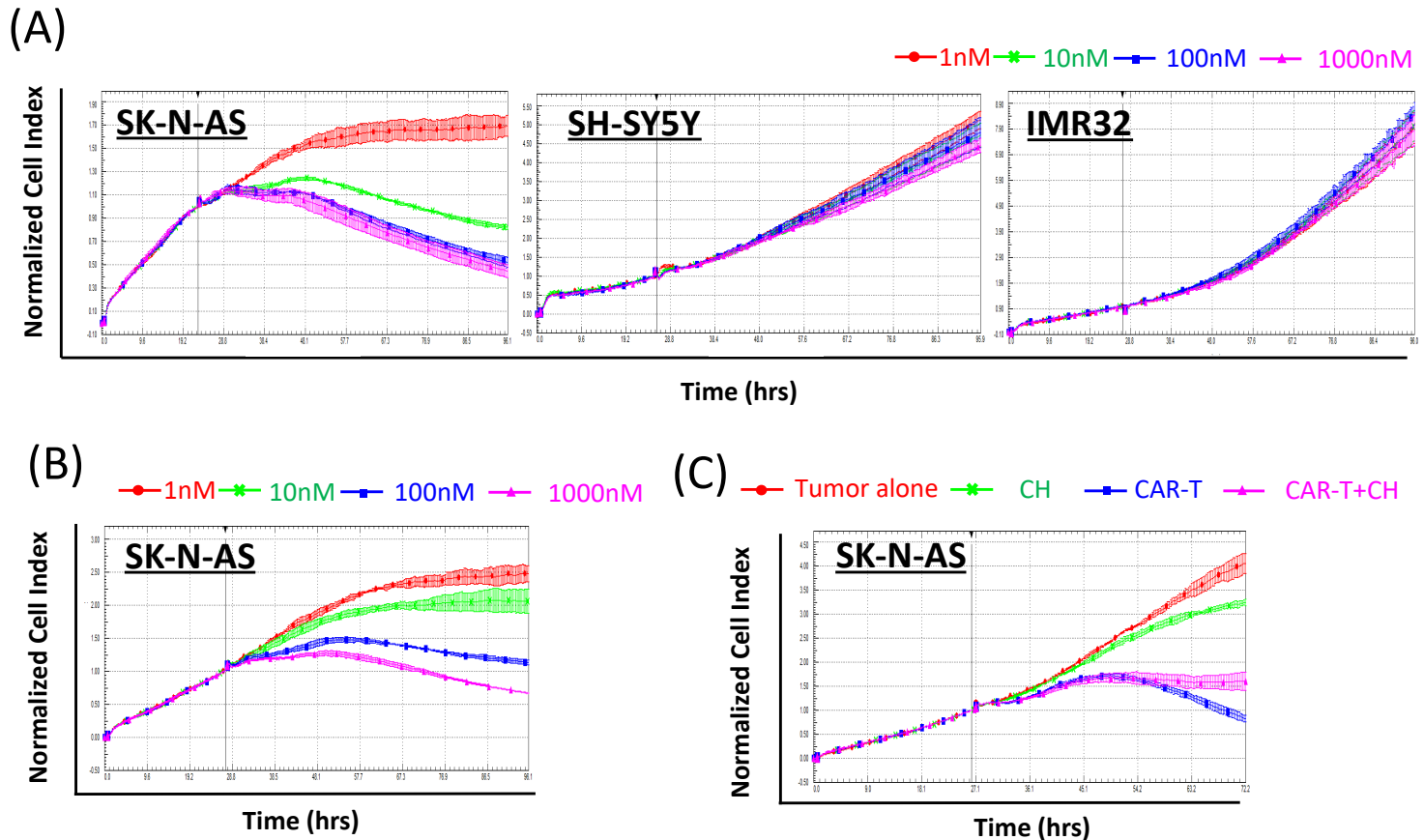


Figure S2 Sensitivity of neuroblastoma (NB) cell lines to MEK inhibitor.

(A) (B) Mutation data for NB cells and an impedance-based cell viability assay. GD2-positive NB cells were seeded on xCELLigence E-plates. After 24 h incubation, vehicle or inhibitors, trametinib (A) or CH5126766 (B) was applied to the cells over a range of concentrations from 1 nM to 1000 nM in triplicate. (C) An impedance-based tumor cell killing assay. SK-N-AS and PB-GD2-CAR-T cells with or without CH5126766 (10 nM), a Raf/MEK inhibitor, were cocultured at an effector:target (E:T) ratio of 2:1. CH; CH5126766.

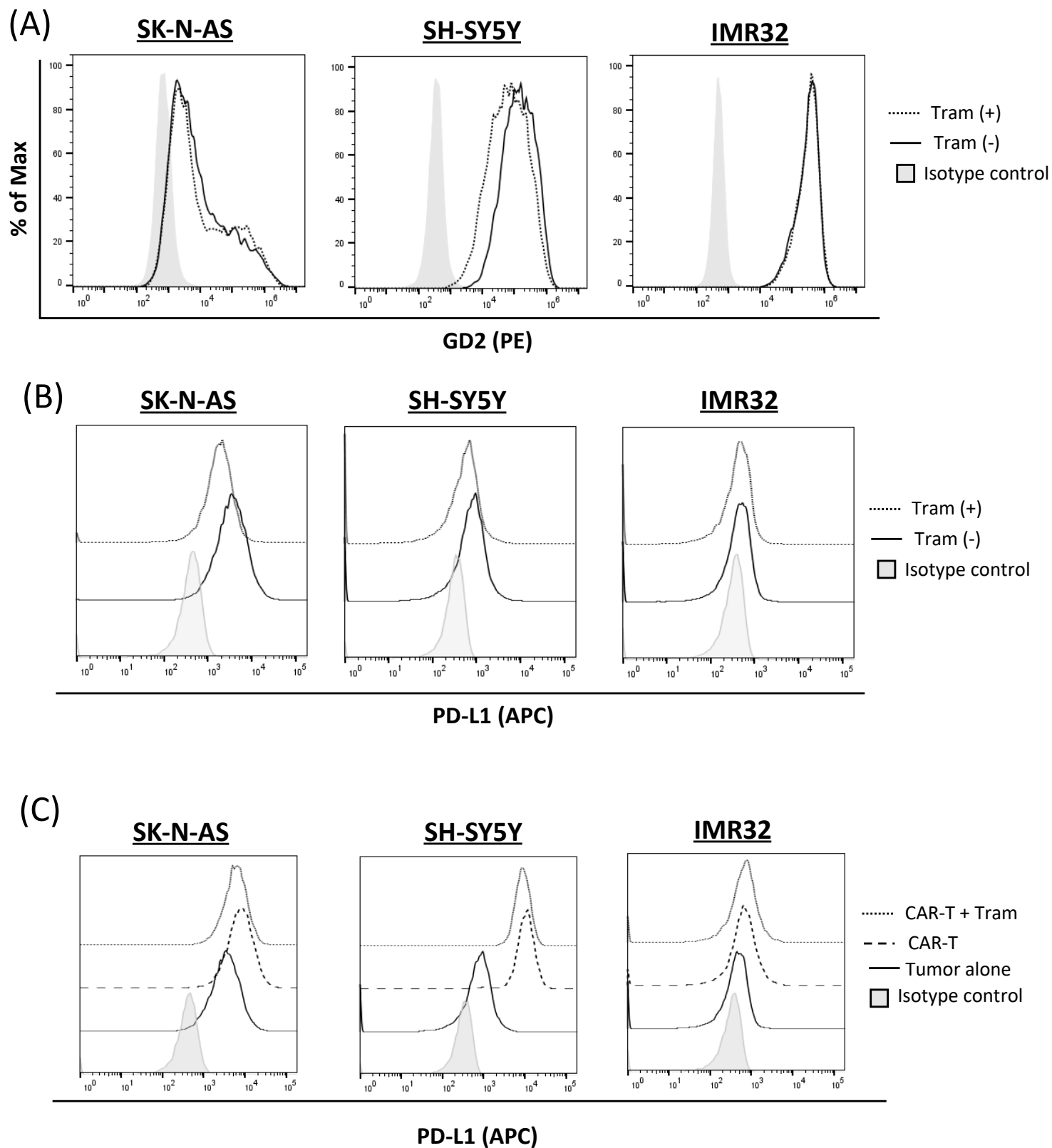
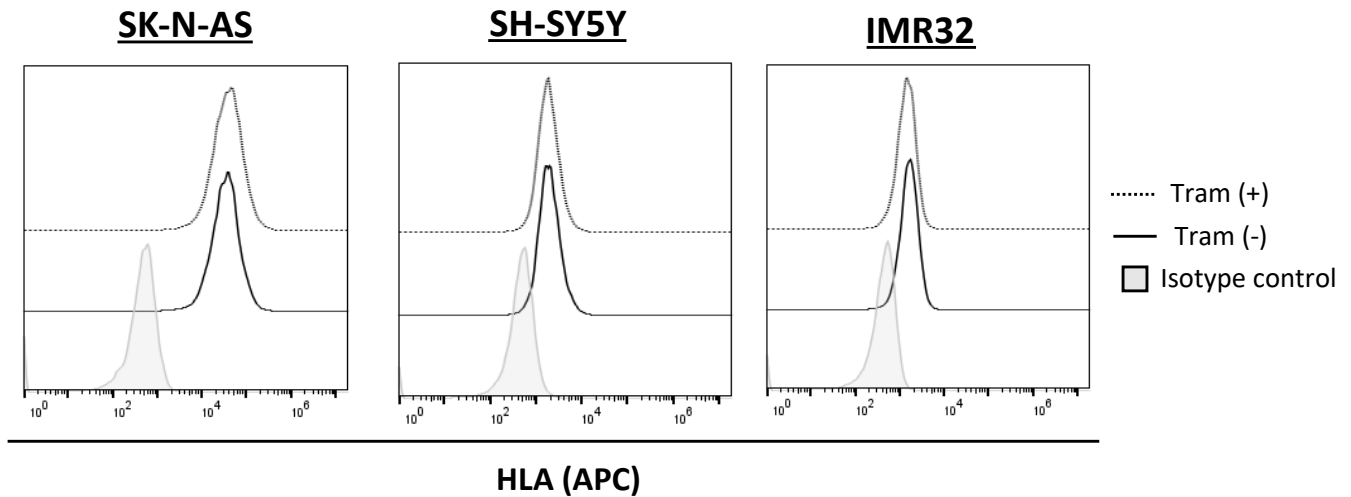


Figure S3 Trametinib did not affect the expression of GD2, and slightly decreased the expression of PD-L1 on neuroblastoma (NB) cells.

(A) (B) Additional effects of trametinib (10 nM) on GD2 (A) or PD-L1 (B) expression on NB cells after 48 h coculture. (C) Additional effects of trametinib on PD-L1 expression on NB cells in coculture with PB-GD2-CAR-T cells for 48 h.

(A)



(B)

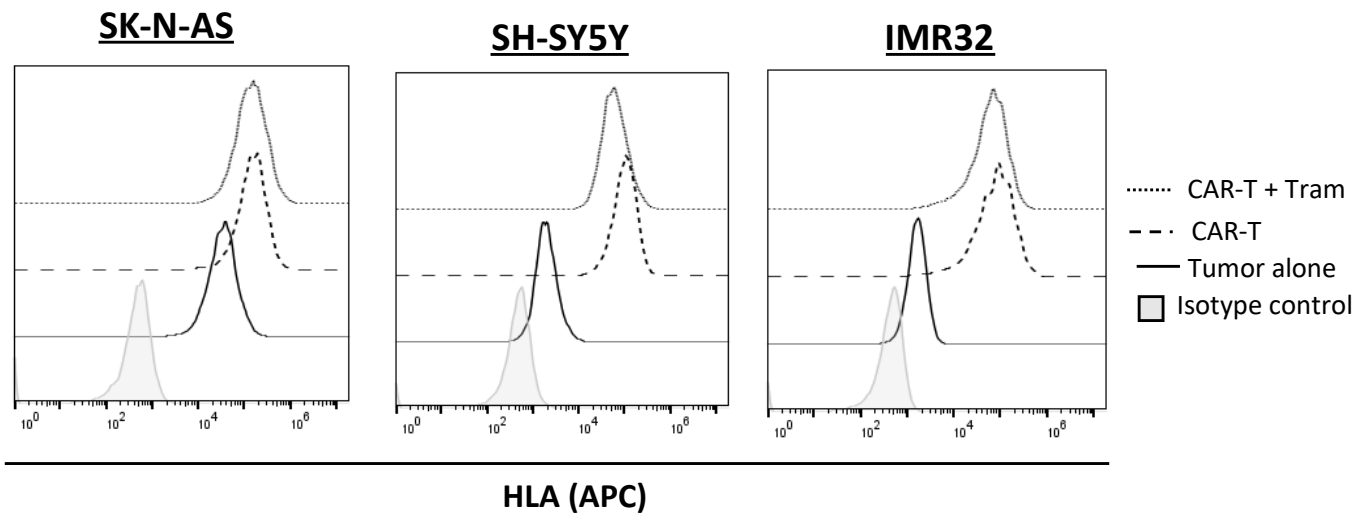


Figure S4 Trametinib did not affect the expression of HLA on neuroblastoma (NB) cells.

(A) Additional effects of trametinib (10 nM) on HLA expression on NB cells after 48 h coculture. (B) Additional effects of trametinib on HLA expression on NB cells in coculture with PB-GD2-CAR-T cells for 48 h.

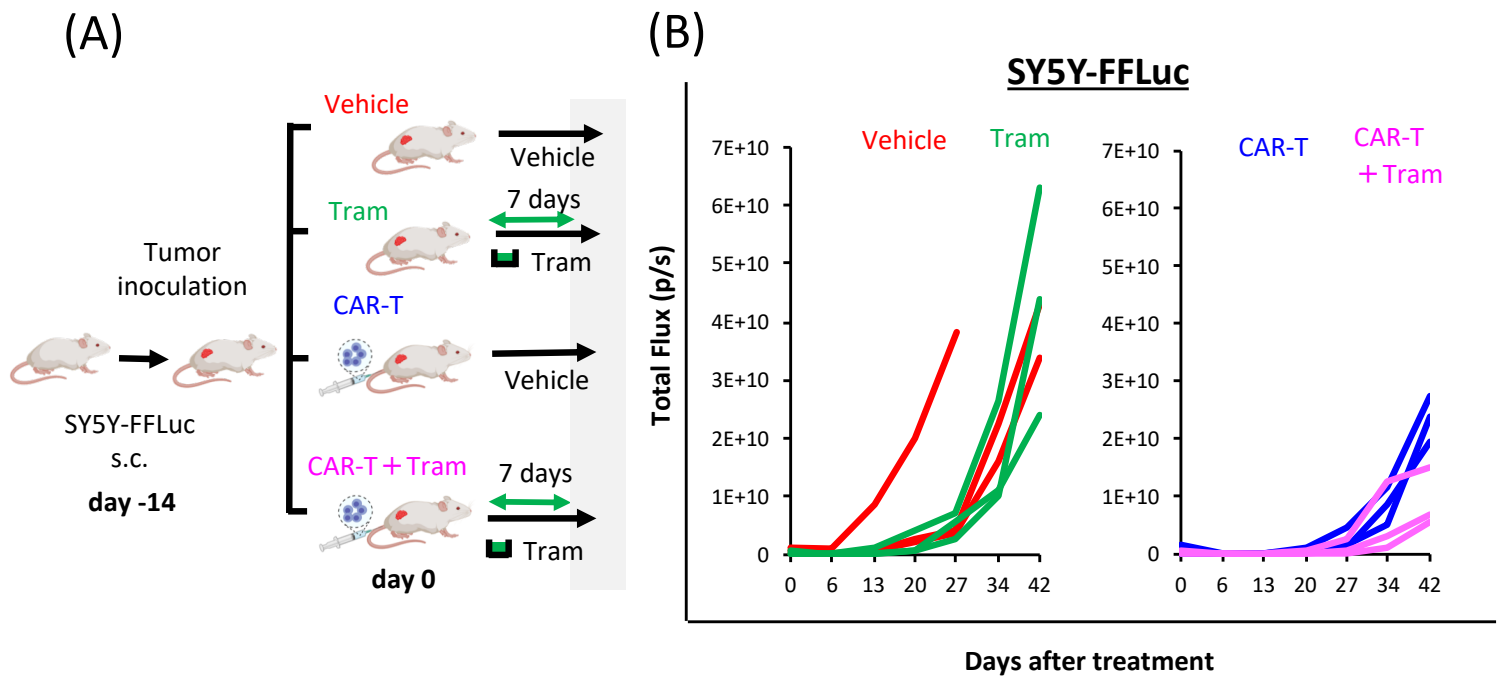
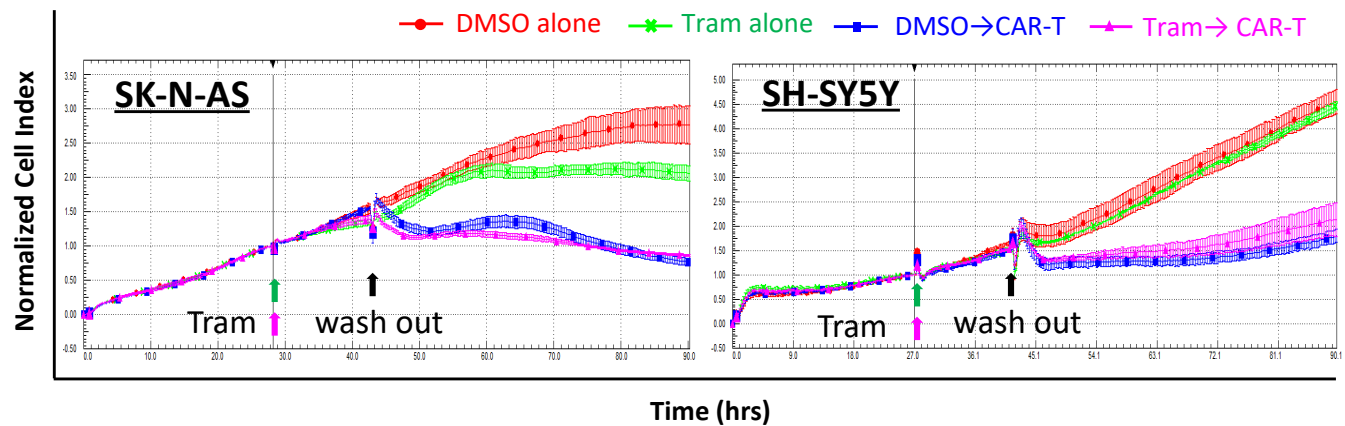


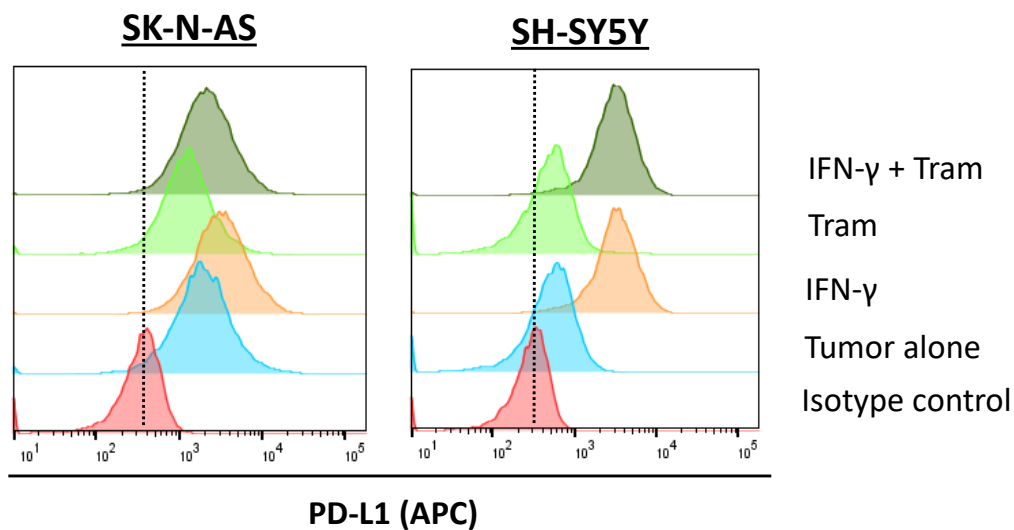
Figure S5 Combinatorial therapy with PB-GD2-CAR-T cells and trametinib in MAPK pathway non-mutated tumor mouse.

(A) Schematics of tumor challenge experiments in the xenograft model. After SY5Y-FFLuc tumor engraftment, mice were divided into four experimental groups (n = 3), as indicated. (B) Tumor growth was measured as bioluminescence signal intensity (BLI) and expressed as total flux (p/s). The combination treatment group demonstrated a reduction in tumor growth, measured as the mean total flux relative to the PB-GD2-CAR-T cell group. Tram; Trametinib.

(A)



(B)



(C)

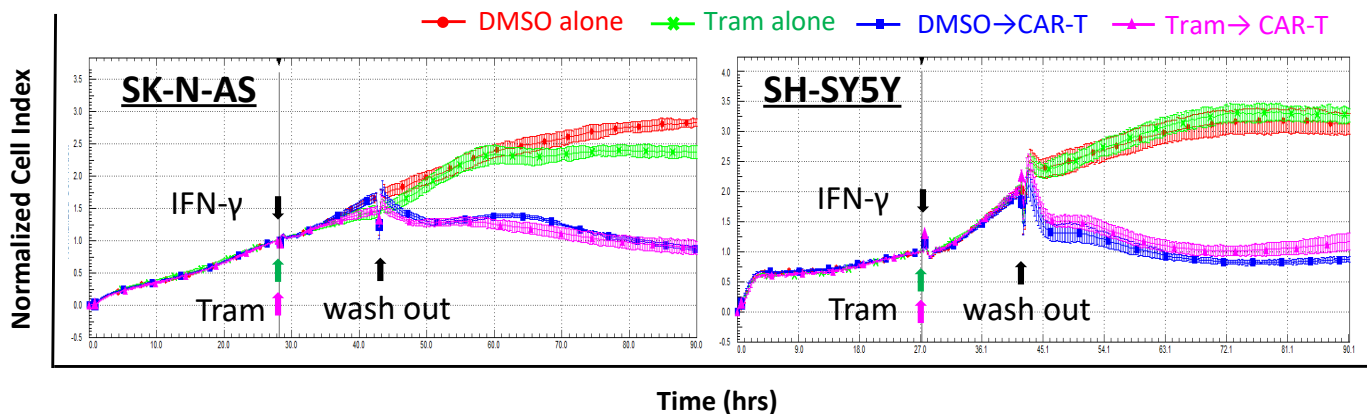


Figure S6 Downregulation of PD-L1 on NB cells by trametinib is poorly involved with the mechanisms of enhanced antitumor effect of PB-GD2-CAR-T cells.

(A) An impedance-based tumor cell killing assay. Approximately 24 h after the seeding of NB cells, trametinib (10 nM) or dimethyl sulfoxide (DMSO, control) was added. Approximately 16 h after adding, trametinib and DMSO was washed out prior to the coculture with PB-GD2-CAR-T cells at an effector:target (E:T) ratio of 2:1 in triplicate. Real-time impedance traces of tumor cells without treatment (DMSO alone; red), and tumor cells cocultured with trametinib (Tram alone; light-green), PB-GD2-CAR-T cells (DMSO→CAR-T; blue) or PB-GD2-CAR-T cells following trametinib (Tram→CAR-T; pink), were acquired for 90 h. (B) Additional effects of trametinib on PD-L1 expression on NB cells in coculture with IFN- γ (10 ng) for 48 h. (C) An impedance-based tumor cell killing assay. Approximately 24 h after the seeding of NB cells, trametinib (10 nM) or DMSO (control) was added in the presence of IFN- γ (10 ng). Approximately 16 h after adding, trametinib, DMSO, and IFN- γ were washed out prior to the coculture with PB-GD2-CAR-T cells at an E:T ratio of 2:1 in triplicate. Real-time impedance traces of tumor cells were acquired for 90 h, as shown in Figure S6A